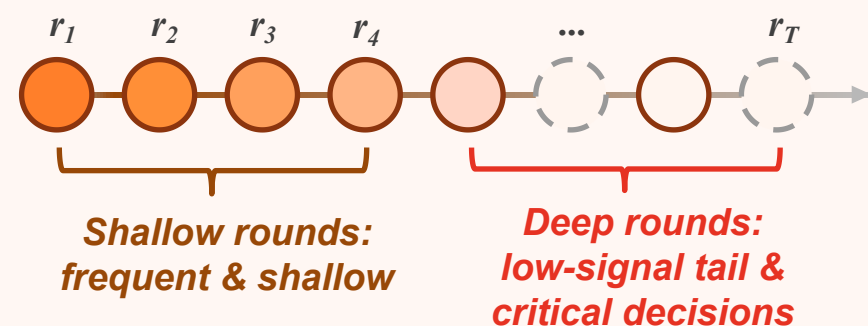


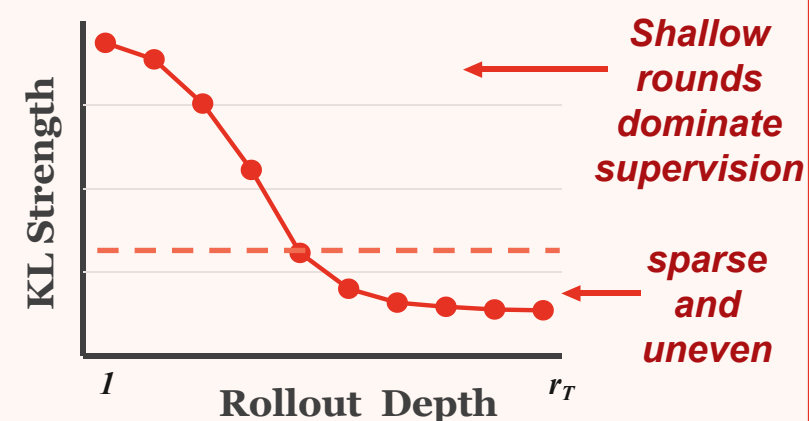
Vanilla OPD

A External mismatch

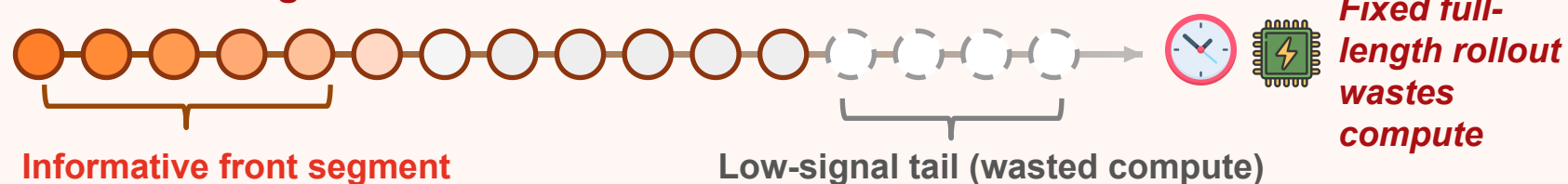
Long-horizon agent trajectory over rounds



Reverse KL signal vs. round depth

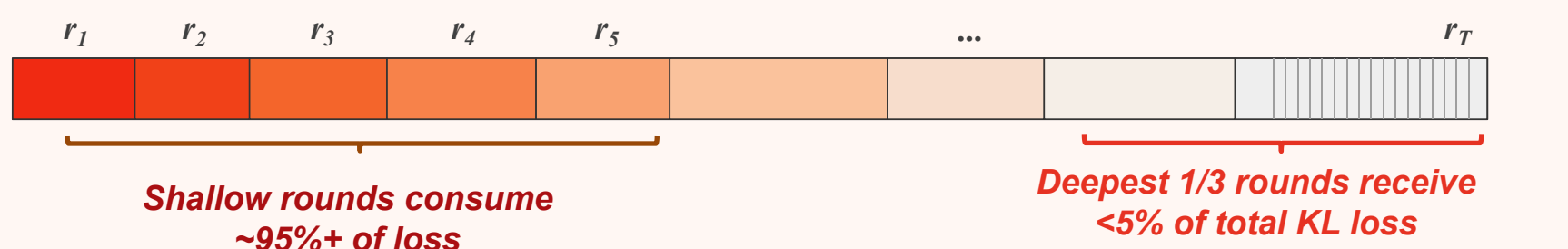


Fixed full-length rollout



B Internal mismatch

(a) Loss allocation under token-level normalization



(b) Naive uniform round weighting

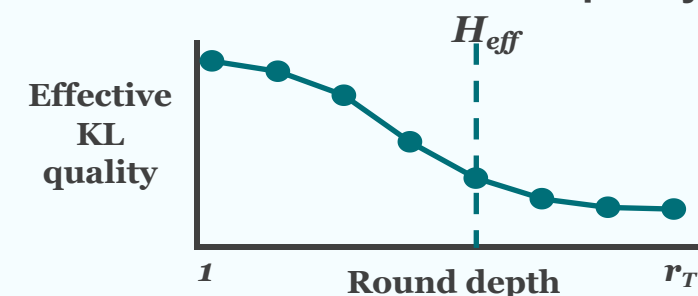
Turn Index	r_1	r_2	r_3	r_4	...	r_T
Weight	1	1	1	1	...	1

Deep turns are weak and rarely observed

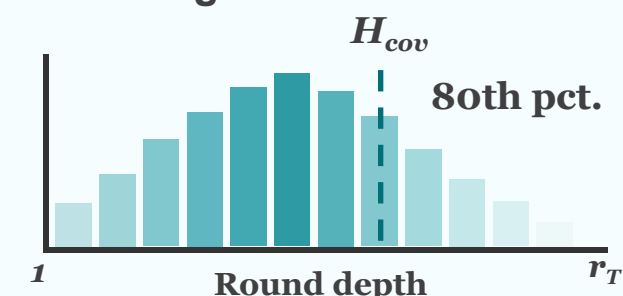
TurnOPD

C Adaptive rollout depth controller

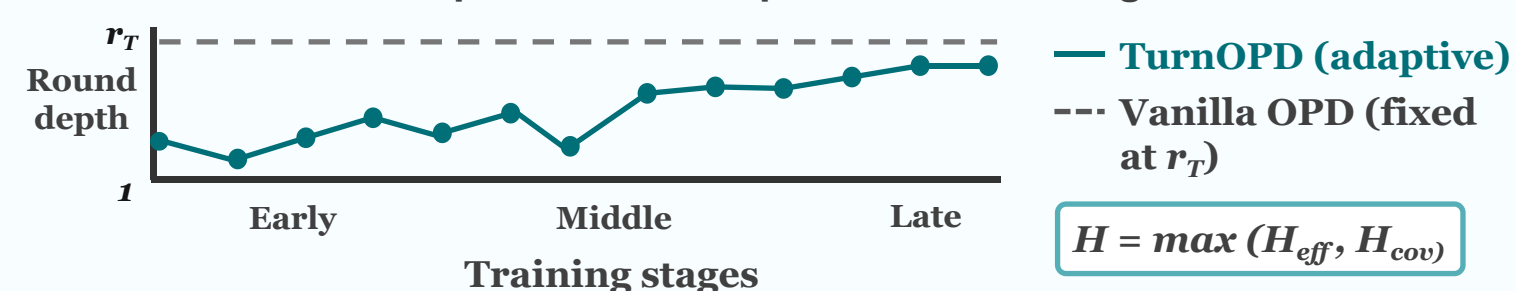
Per-round effective KL quality



Coverage Side Lower-Bound

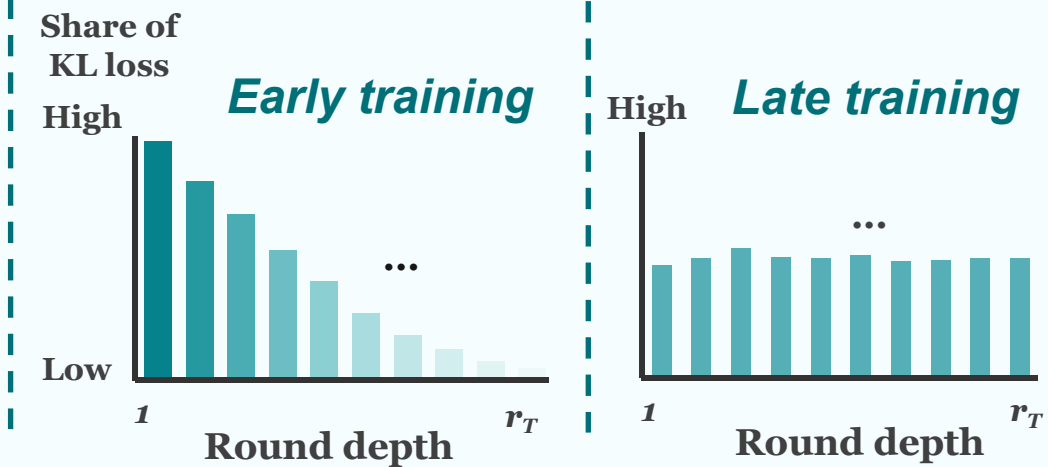
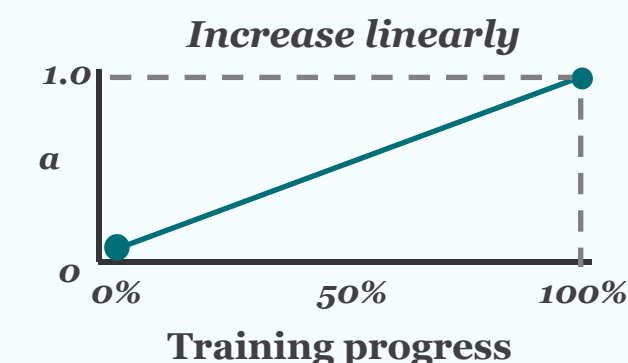


Adaptive rollout depth across training



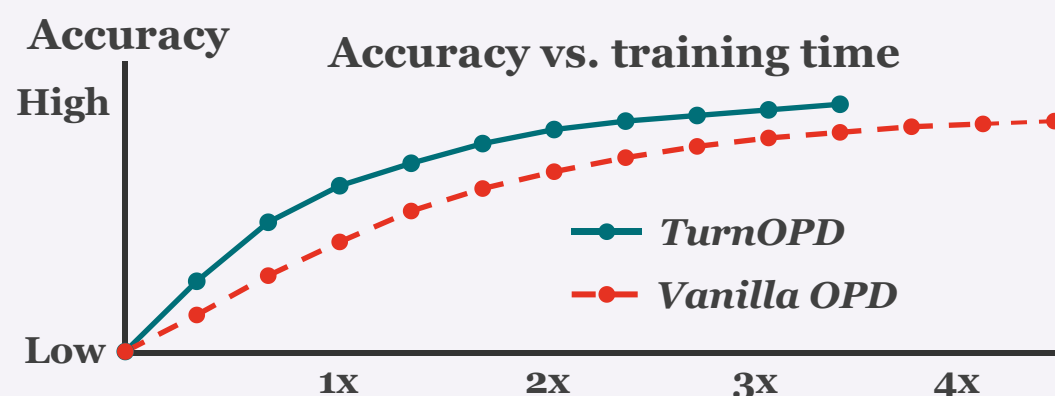
D Linear round-normalization mixing

$$L_{mix} = (1 - \alpha)L_{token} + \alpha L_{round}$$



Choose **how many** rounds to collect

Choose **how** to allocate loss



Higher minimum
time-averaged accuracy



Up to 2.29x
faster training